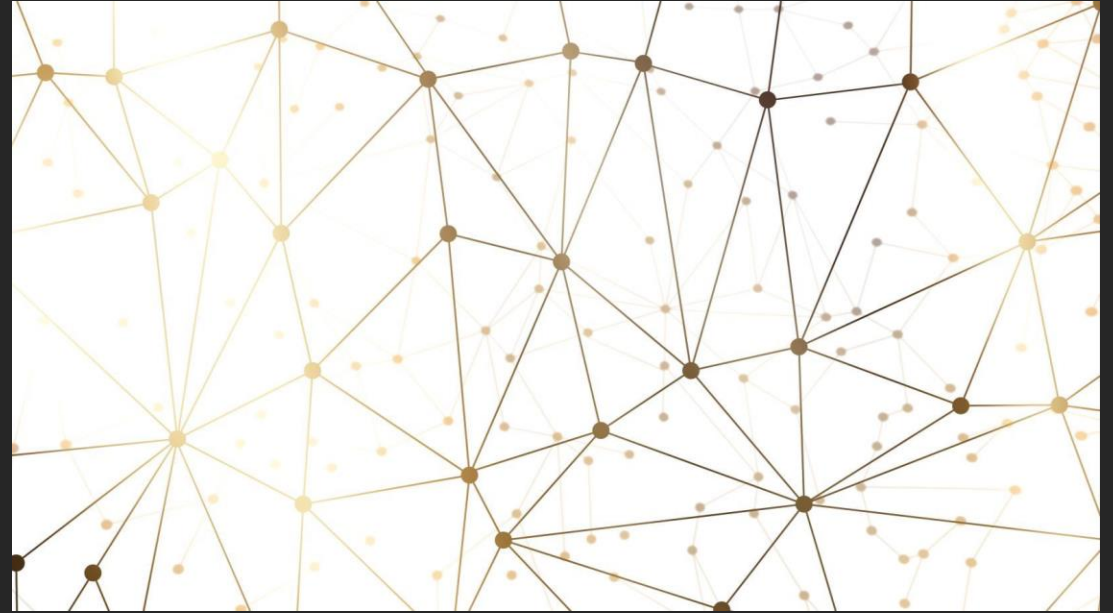


Preprocessing Training

QIUYU YU



Load Modules on HPC

```
#!/bin/bash

# extract your ID
echo "$USER"

ml Flask/2.3.3-GCCcore-12.3.0
ml netpbm/10.73.43-GCC-12.3.0
ml AFNI/24.3.06-foss-2023a

# create working directory
mkdir -p /scratch/${USER}/workDir/rawdata

# copy scripts
cp -r
/work/cglab/projects/BRANCH/all_data/for_AF
NI/processing_scripts/script_for_practice
/home/${USER}/Desktop/script_for_practice
```

Download Files



Files and Applications

Downloads

Save files to

/scratch/qy49547/workDir/rawdata

Browse...

☐ Always ask you where to save files

Run Scripts

- **In terminal:**

`chmod +x script.sh` # give execute access

`./script.sh` # *run script in the current folder*

`bash path/script.sh`

`source script.sh` # *Only if script.sh sets environment variables or loads modules*

`python script.py`

Submit a job:

`sbatch script.sh`

Cancel a job:

`scancel jobID`

Unzip DICOM Files and Reconstruction

```
#!/bin/bash
#SBATCH --job-name=unzip_recon          # Job name
#SBATCH --partition=batch              # Partition (queue) name
#SBATCH --ntasks=1                    # Run a single task
#SBATCH --cpus-per-task=10             # Number of CPU cores per task
#SBATCH --mem=80gb                     # Job memory request
#SBATCH --time=04:00:00                # Time limit hrs:min:sec
#SBATCH --output=/scratch/%u/workDir/log/unzip_recon_%A-%a.out
#SBATCH --error=/scratch/%u/workDir/log/unzip_recon_%A-%a.err

# Define base_dir before using it
export myID="$USER"
export base_dir="/scratch/$myID/workDir"
export my_scripts="/home/$myID/Desktop/scripts"
export input_dir="${base_dir}/rawdata"

# Load modules
ml Flask/2.3.3-GCCcore-12.3.0
ml netpbm/10.73.43-GCC-12.3.0
ml AFNI/24.3.06-foss-2023a
ml p7zip/17.04-GCCcore-11.3.0
ml parallel/20240322-GCCcore-13.2.0

time "$my_scripts/unzip_scratch.sh"
time "$my_scripts/convert_dicom_scratch.sh"
```



Data Clean

Data Clean

You will never know what kind of issue you will encounter when you do the data cleaning.

Objective: All the files have the same naming format. All participants' folders have the same folder structure. Remove unnecessary files. The raw directory will keep all NIfTI files that are reconstructed from raw DICOM files, and the derivatives folders will have a copy of the corresponding files.

1. Clean up unnecessary files

- Hidden files: Some systems (e.g. Macs) automatically carry hidden files that will fail to be renamed if you don't clean them up and run the rename script.
- Duplicate files
 - Extra copy.
 - Two T1, two T2: keep the better one to analyze in the derivatives folder and remove the extra one (but keep one in the raw directory).
- Unnecessary derivative files that were generated by some other software.
- Remove additional DICOM files.

1. Rename

- Part of the NIfTIs files that were reconstructed from DICOM files may have systematic naming issues (may need to be completely renamed).
- filed map (PM) files
 - name them to PM1_4 and PM2_5. The echo time should match the file name, which means 1-4 should be the shorter echo (0.005s) and 2-5 should be the longer echo (0.0072s).
 - There should be 8 field map files in total, 2 phase maps and 2 magnitude maps, and 1 corresponding JSON file for each.
- Rename fields
 - BRANCH, Brch, etc. to BRCH

Slurm Scripts: Tree (trunk branch), and Forest



Slurm Scripts: Tree (trunk, branch), and Forest



```
#!/bin/bash
#SBATCH --job-name=afni_preprocessing      # Job name
#SBATCH --partition=batch                  # Partition (queue) name
#SBATCH --nodes=1
#SBATCH --ntasks=1                        # Run a single task
#SBATCH --cpus-per-task=20                 # Number of CPU cores per task
#SBATCH --mem-per-cpu=1gb
#SBATCH --time=23:00:00                    # Time limit hrs:min:sec
#SBATCH --output=/scratch/%u/workDir/log/preprocessing/afni_%A-%a.out
#SBATCH --error=/scratch/%u/workDir/log/preprocessing/afni_%A-%a.err
#SBATCH --array=1-6%20                     # Array range: DONT FORGET TO CHANGE!!!
                                         placeholder = myID

# Define base_dir before using it
# myID=$(echo "$(pwd)" | cut -d'/' -f3)
export myID="$USER"
export base_dir="/scratch/$myID/workDir"
export scripts_dir="/work/cglab/projects/BRANCH/all_data/for_AFNI/processing_scripts"
export my_scripts="/home/$myID/Desktop/script_for_practice"

# Load AFNI module
ml Flask/2.3.3-GCCcore-12.3.0
ml netpbm/10.73.43-GCC-12.3.0
ml AFNI/24.3.06-foss-2023a

#ml FSL/6.0.7.14-foss-2023a
#source ${FSLDIR}/etc/fslconf/fsl.sh

subjects=(163 182 184 187 189 192) 6
subject=${subjects[${SLURM_ARRAY_TASK_ID} - 1]}

#time "$my_scripts/sswarp_scratch_main.sh" ${subject}
#time "$my_scripts/bias_field_correction_scratch_v2.0.sh" ${subject}
```


Main scripts

pass the
legacy from
slurm scripts

```
#!/bin/bash

subj="$(printf %03d $1)"

dir_path="${base_dir}/${subj}/derivatives/sswarp2/"

export AFNI_NO_X11=1

# Check if the specified path is a valid directory
if [[ ! -d "$dir_path" ]]; then
    echo "wrong ID"
    exit 1
fi

cd "$dir_path" || exit 1

# create T1_results folder
mkdir -p "$dir_path/T1_results"

for nifti_file in BR*.nii.gz; do
    if [ -f "$nifti_file" ]; then
        echo "processing: $nifti_file"
        echo "ID: $subj"

        sswarper2 \
            -input "$nifti_file" \
```

```
#!/bin/bash
echo "$1"
subj="$(printf %03d $1)"

cd "${base_dir}/${subj}/derivatives/kidvid_output"

export AFNI_NO_X11=1    mute warning

#####
### GET THE CORRECT FILE NAME FOR THE DATA
# Find the file with 'FCEV' and '.nii' in its name
file_name=$(find . -type f -name '*CEV*.nii*' -print -quit)
# Check if the file was found
if [ -z "$file_name" ]; then
    echo "No file with 'CEV' and '.nii' in its name found."
    exit 1
else
    echo "File found: $file_name"
fi

## file name is now stored as $file_name
### ASSIGN CORRECT FILE NAME TO 'rawdata' VARIABLE#####
export rawdata="$file_name" #CHANGE THIS
#####
### Run afni_proc.py to create a single subject processing script (proc.py) and execute it
afni_proc.py \
    -subj_id $subj \
    -script proc."${subj}" -scr_overwrite \
    -out_dir "${subj}.results" \
    -copy_anat "${base_dir}/${subj}/derivatives/sswarp2/T1_results/anatSS."${subj}".nii \
    -anat_has_skull no \
    -dsets $rawdata \
    -blocks tshift align tlrc volreg blur mask scale regress \
    -radial_correlate_blocks tcat volreg regress \
    -tcat_remove_first_trs 6 \
```

Errors

[afni_37276104-0.out](#)

[afni_37276120-15.err](#)

```
-- trying to start Xvfb :158
[1] 1334245
The XKEYBOARD keymap compiler (xkbcomp) reports:
> Warning:      Could not resolve keysym XF86AudioPreset
> Warning:      Could not resolve keysym XF86Keyboard
> Warning:      Could not resolve keysym XF86MonBrightnessCycle
> Warning:      Could not resolve keysym XF86BrightnessAuto
> Warning:      Could not resolve keysym XF86DisplayOff
> Warning:      Could not resolve keysym XF86WWAN
> Warning:      Could not resolve keysym XF86RFKill
> Warning:      Could not resolve keysym XF86Info
> Warning:      Could not resolve keysym XF86FullScreen
> Warning:      Could not resolve keysym XF86AspectRatio
> Warning:      Could not resolve keysym XF86DVD
> Warning:      Could not resolve keysym XF86Audio
> Warning:      Could not resolve keysym XF86ChannelUp
> Warning:      Could not resolve keysym XF86ChannelDown
> Warning:      Could not resolve keysym XF86Break
> Warning:      Could not resolve keysym XF86VideoPhone
> Warning:      Could not resolve keysym XF86ZoomReset
> Warning:      Could not resolve keysym XF86Editor
> Warning:      Could not resolve keysym XF86GraphicsEditor
```

Check Missing Subjects

```
qy49547@c5-12 organize_scripts$ ./check_results.sh
```

```
T1 analysis (0 subjects missing)
```

```
kidvid A analysis (0 subjects missing)
```

```
kidvid B analysis (1 subjects missing)
```

```
132
```

```
kidvid_other (1 subjects missing)
```

```
044
```

ID	T1	kidvid	kidvid_bf	path						
s001	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s001/						
s003	1	NA	NA	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s003/						
s004	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s004/						
s005	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s005/						
s007	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s007/						
s008	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s008/						
s009	1	NA	NA	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s009/						
s010	1	NA	NA	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s010/						
s012	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s012/						
s013	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s013/						
s016	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s016/						
s017	1	1	0	/work/cglab/projects/BRANCH/all_data/for_AFNI/BRANCH_W1/s017/						

Extra Step: rename and tar DICOM files

The screenshot displays a File Manager interface with two panels. The left panel shows a collection of files in a directory, and the right panel shows a list of files with two items selected.

Left Panel:

- Collection: UGA GACRC Collection (All GACRC filesystems) - GACRC Users only
- Path: /project/cglab/BRANCH/DataTransfers/BRANCH_downloads/
- Buttons: Start, Transfer & Timer Options, Start
- Actions: select all, up one folder, refresh list, filter, view
- Table:

NAME	LAST MODIFIED	SIZE
backup	5/20/2025, 12:12 PM	—
s174.tar.gz	5/5/2025, 09:26 PM	1.86 GB
s181.tar.gz	5/5/2025, 09:26 PM	1.86 GB
download_2025-04-21_16-30-17.zip	4/22/2025, 11:34 AM	4.48 GB
download_2025-04-21_16-29-55.zip	4/22/2025, 11:34 AM	4.50 GB
download_2025-04-21_16-30-07.zip	4/22/2025, 11:34 AM	4.31 GB
s187.tar.gz	4/22/2025, 11:34 AM	1.66 GB
s189.tar.gz	4/22/2025, 11:34 AM	1.59 GB
s188.tar.gz	4/22/2025, 11:34 AM	1.52 GB
s177.tar.gz	4/21/2025, 04:42 PM	1.32 GB
s192.tar.gz	4/21/2025, 04:42 PM	1.33 GB
s184.tar.gz	4/21/2025, 04:42 PM	1.59 GB

Right Panel:

- Collection: UGA GACRC Collection (All GACRC filesystems) - GACRC Users only
- Path: /home/qy49547/zip_all_files/
- Buttons: Start
- Actions: select none, up one folder, refresh list, filter, view
- Table:

NAME	LAST MODIFIED	SIZE
☒ s200.tar.gz	5/20/2025, 12:53 PM	1.86 GB
☒ s202.tar.gz	5/20/2025, 12:49 PM	133.56 MB
☐ dicom_summary_2025-05-20_12-49-23.csv	5/20/2025, 12:49 PM	192 B
☐ rename_log_2025-05-20_12-49-23.txt	5/20/2025, 12:49 PM	88 B
☐ s200	5/20/2025, 11:43 AM	—
☐ s202	5/14/2025, 06:25 PM	—
☐ dicom_summary_2025-05-05_13-38-43.csv	5/5/2025, 01:38 PM	192 B
☐ rename_log_2025-05-05_13-38-43.txt	5/5/2025, 01:38 PM	88 B
☐ dicom_summary_2025-04-22_11-15-35.csv	4/22/2025, 11:15 AM	265 B
☐ rename_log_2025-04-22_11-15-36.txt	4/22/2025, 11:15 AM	110 B
☐ dicom_summary_2025-04-21_16-29-48.csv	4/21/2025, 04:29 PM	346 B

2 items selected

Extra Step: rename and tar DICOM files

```
Renaming Log - Mon Apr 21 16:29:49 EDT 2025  
Renamed: e217 -> s170  
Renamed: e218 -> s177  
Renamed: e220 -> s184  
Renamed: e221 -> s192
```

e_Folder	ID	Branch_N	File	File Path			
e217	BRANCH_	s170	i12398428	e217/s2231/i12398428.MRDC.21			
e218	BRANCH_	s177	i12398538	e218/s2233/i12398538.MRDC.11			
e220	BRANCH_	s184	i12665721	e220/s2256/i12665721.MRDC.44			
e221	BRANCH_	s192	i12781361	e221/s2269/i12781361.MRDC.11			